LV 2 - LR gramatika

Grupa 8

SelectStatement → select begin CaseList end
CaseList → CaseList Case | Case
Case → case ID => Statement
Statement → SelectStatement | ID:= ID; | ID:= CONST;

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1) SS → select begin CL end
2) CL → CL C
3) CL → C
4) C → Case ID => S
5) S → SS
6) S → ID:=ID;
7) S → ID:=CONST;

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1. Kanonični skup LR pravila

Uvodimo smenu I_0: SS' → . SS

SS je neterminal:

SS → . select begin CL end

prelazi: I_1=goto(I_0, SS), I_2=goto(I_0, select)

I_1: SS' → SS. (nema redukcije jer je fiktivna smena)

I_2: SS → select . begin CL end

prelazi: I_3=goto(I_2, begin)

I_3:

SS → select begin . CL end

CL → . CL C

CL → . C

C → case ID => S

prelazi: I_4=goto(I_3, CL), I_5=goto(I_3, C), I_6=goto(I_3, case)

I_4 $SS \to \underset{select}{\lambda}$ begin CL . end

$CL \to CL . C$

$C \to .$ case $ID \Rightarrow S$

prelazi: $I_7 = goto(I_4, end)$, $I_8 = goto(I_4, C)$, $I_9 = goto(I_4, case)$

I_5 $CL \to C.$ Reduciraio stanje za smenu 3

I_6 $C \to case . ID \Rightarrow S$

prelazi $I_9 = goto(I_6, ID)$

I_7 $SS \to$ select begin CL end. - Reduciraio stanje za smenu 1

I_8 $CL \to CL\ C.$ - Reduciraio stanje za smenu 2

I_9 $C \to case\ ID . \Rightarrow S$

$I_{10} = goto(ID, \Rightarrow)$

I_{10} $C \to case\ ID \Rightarrow . S$

$S \to . SS$ $\quad$ $SS \to .$ ~~begin~~ select begin CL end

$S \to . ID := ID;$

$S \to . ID := CONST;$

$I_{11} = goto(I_{10}, S)$, $I_{12} = goto(I_{10}, SS)$ $I_{13} = goto(I_{10}, ID)$

$I_2 = goto(I_{10}, select)$

I_{11} $C \to case\ ID \Rightarrow S.$ - Reduciraio stanje za smenu 4

I_{12} $S \to SS.$ - Reduciraio stanje za smenu 5

I_{13} $S \to ID := ID;$

$S \to ID . := CONST;$

$I_{14} = goto(I_{13}, :=)$

I_{14} $S \to ID := . ID;$

$S \to ID := . CONST;$

$I_{15} = goto(I_{14}, ID)$ $\quad$ $I_{16} = (I_{14}, CONST)$

CamScanner

$S \to ID := ID \cdot ;$

$I_{17} = goto(I_{15}, ;)$

I_{16}

$S \to ID := CONST \cdot ;$

$I_{18} = goto(I_{16}, ;)$

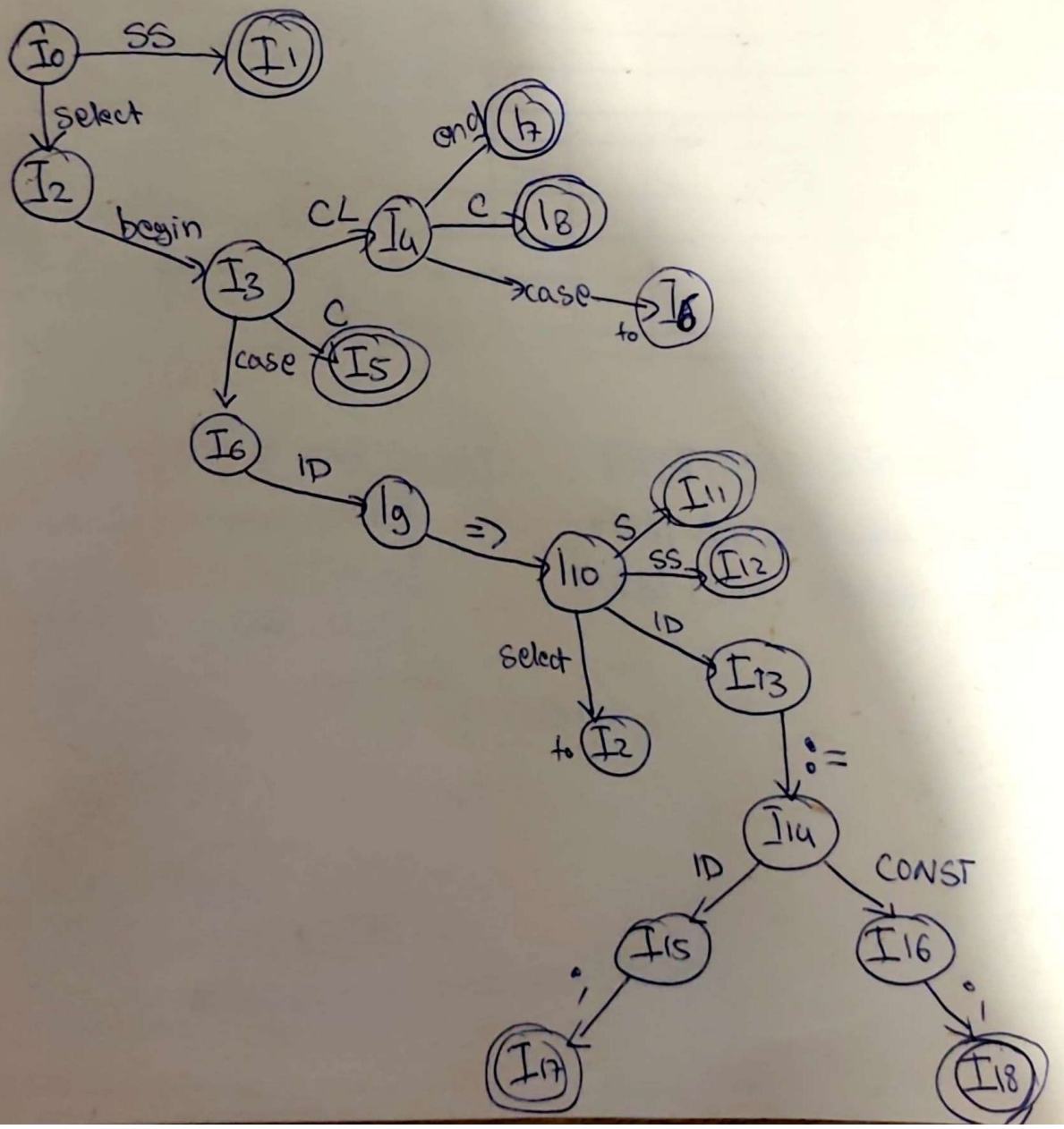
I_{17}

$S \to ID := ID ; \cdot$ - Redukcijo stanje za smenu 6

I_{18}

$S \to ID := const ; \cdot$ - Redukcijo stanje za smenu 7

2. Graf prelaza



Popunjavanje LR sintaksne tabele

	;	select	begin	end	case	ID	=>	:=	CONST	#	SS	CL	S	C
0		s2									1			
1														
2			s3											
3					s6							4		5
4				s7	s6									8
5				r3	r3									
6						s9								
7				r1	r1					r1				
8				r2	r2									
9							s10							
10						s13					12		11	
11				r4	r4									
12				r5	r5									
13								s14						
14						s15			s16					
15	s17													
16	s18													
17				r6	r6									
18				r7	r7									

FOLLOW(SS) = {#, end, case}
FOLLOW(CL) = {end, case}
FOLLOW(C) = {end, case}
FOLLOW(S) = {end, case}

FIRST(SS) = {select}
FIRST(CL) = {case}
FIRST(S) = {select, ID}

$I_0 \xrightarrow{SS} I_1$ → neterminal → pišemo 1

$I_0 \xrightarrow{select} I_2$ → s2

$I_3 \xrightarrow{C} I_5$

I_5: CL → C•, red. 3

FOLLOW(CL) = {end, case} => r3